The Intuitive Appeal of Explainable Machines (Selbst & Barocas 2018)

Legal and regulatory demands for algorithmic explainability conflate inscrutability (epistemic opacity) with non-intuitiveness (statistical counter-intuitiveness), creating governance gaps when applied to complex models.

GRADE: HIGH | STATUS: STABLE | DOMAIN: ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Regulators and legal scholars demand "explanation" for automated decisions, assuming machine explanations can serve the same function as human reasons (justification, contesting decisions, ensuring non-discrimination). Selbst & Barocas demonstrate that this demand conflates two distinct forms of opacity: inscrutability (inability to inspect the mathematical path from input to output) and non-intuitiveness (the statistical logic operates on high-dimensional correlations that defy human causal heuristics).

EVIDENCE AND CAUSAL ASSESSMENT

Even if a deep neural network's calculations are rendered fully transparent (complete mathematical inspectability), its statistical rules frequently remain unintelligible to human operators because the model relies on weak, non-linear correlations across millions of features that cannot be translated into traditional normative reasons.

EFFECT SIZE AND PERSISTENCE

Post-hoc explainability techniques (e.g., LIME, SHAP, feature saliency) provide local linear approximations that can be misleading or easily gamed, failing to satisfy the legal standard of substantive causal accountability.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: n/a.

Classified as robust (<20%). Demands for simplified intuitive explanations frequently force organizations to discard predictive accuracy or resort to superficial explanations that conceal underlying disparate impacts.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

The difficulty of intuitive machine explanation does not excuse organizations from compliance; rather, it indicates that regulation must focus on empirical outcome testing and system constraints rather than post-hoc verbal justifications.

WHAT ACTION IS REASONABLY JUSTIFIED

In regulated industries, ground compliance architectures in deterministic constraint layers, auditable retrieval databases, and statistical impact audits rather than relying on LLM-generated narrative self-explanations.

VERIFIED SOURCES & DOI IDENTIFIERS

- Selbst, A. D., & Barocas, S. (2018). The Intuitive Appeal of Explainable Machines. Fordham Law Review, 87(3), 1085–1139. https://doi.org/10.2139/ssrn.3126971